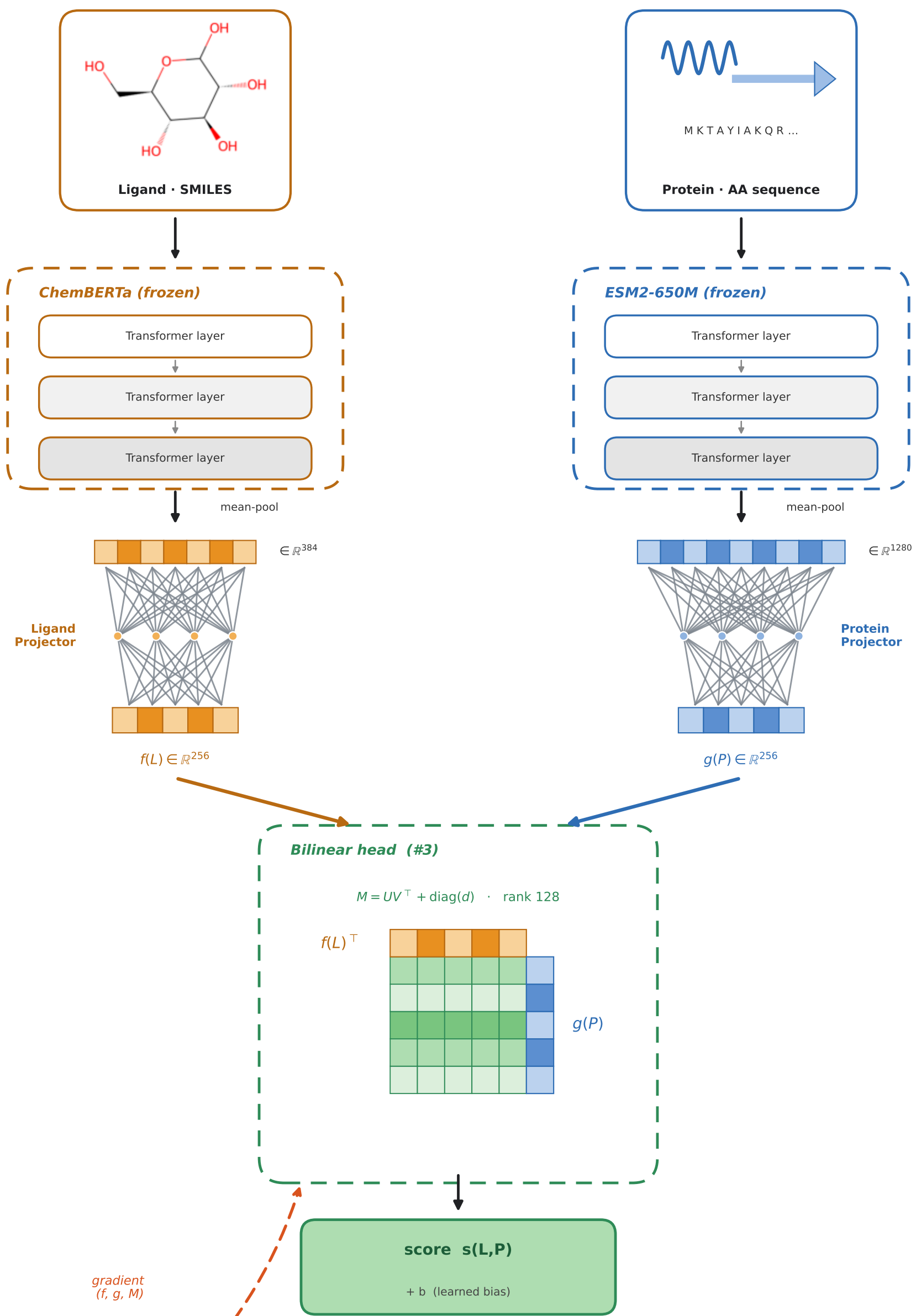


RankBind (v5): model architecture



Anti-shortcut training (margin objective)

- #1 protein-balanced sampling
- #2 within-ligand margin loss
- #4 hard-negative mining (top-50 confusers, refreshed each epoch)